



HNG RIDE_HAILING BUSINESS ANALYSIS REPORT



Explanatory Data Analysis (EDA)

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Date: October 2025

1. Data Cleaning

Before analysis, the dataset underwent the following cleaning steps:

- Column Standardization: Renamed all columns to consistent, descriptive formats.
- Data Type Conversion: Converted date/time and numeric columns to proper types.
- Handling Missing Values: Removed rides with missing critical information.
- Duplicate Removal: Removed duplicate ride entries and repeated payments.
- Data Consistency Checks: Verified ride_id matches across tables and status values.

2. Exploratory Data Analysis (EDA) Summary

- Operational Performance: Stable after 2022 growth peak.
- Payment Adoption: Balanced between cash and digital.
- Cancellation Rate: Moderate (~18%), requires monitoring.
- Rider Satisfaction: High overall (avg ~4.47), top cities Chicago and New York (~4.52).

3. Business Questions, Objectives, Insights & Findings

Question 1: Top 10 Longest Rides

Objective: Identify the longest rides completed across all cities and payment methods to uncover patterns in long-distance travel and fare potential.

Findings & Insights:

- Longest rides were 30.00 km, e.g., Driver_1774 (Calgary → Los Angeles) and Driver_1886 (Ottawa → Vancouver).
- Most top rides occurred between Calgary, Ottawa, Toronto, Boston, and New York.
- Payment was predominantly cash; longest ride paid by voucher.

Query

Query History

```
1  /* Question 1. Find the top 10 longest rides (by distance), including driver name,
2  rider name, pickup/dropoff cities, and payment method.*/
3  SELECT
4      r."Rider_ID",
5      d."Driver_Name",
6      rd."Rider_Name",
7      r."Pickup_City",
8      r."Dropoff_City",
9      r."Distance_KM",
10     p."Payment_Method"
11 FROM rides_raw r
12 JOIN drivers_raw d ON r."Driver_ID" = d."Driver_ID"
13 JOIN riders_raw rd ON r."Rider_ID" = rd."Rider_ID"
14 JOIN payment_raw p ON r."Ride_ID" = p."Ride_ID"
15 ORDER BY r."Distance_KM" DESC
16 LIMIT 10;
```

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	Rider_ID integer	Driver_Name character varying (100)	Rider_Name character varying (100)	Pickup_City character varying (100)	Dropoff_City character varying (100)	Distance_KM numeric (10,2)	Payment_Method character varying (50)
1	292	Driver_1774	Rider_292	Calgary	Los Angeles	30.00	Voucher
2	6131	Driver_1886	Rider_6131	Ottawa	Vancouver	30.00	Voucher
3	8405	Driver_1000	Rider_8405	Chicago	San Francisco	29.99	PayPal
4	3294	Driver_1406	Rider_3294	New York	Toronto	29.99	Cash
Total rows: 10		Query complete 00:00:01.774		CRLF Ln 16, Col 10			

Question 2: Riders Signed Up in 2021 Who Rode in 2024

Objective: Measure long-term rider retention from the 2021 onboarding cohort.

Findings & Insights:

- 1,848 riders from 2021 continued using the platform in 2024.
- Indicates strong retention and loyalty over a 3-year period.

Query Query History

```
1 -- Question 2: How many riders who signed up in 2021 still took rides in 2024?
2 SELECT COUNT(DISTINCT r."Rider_ID") AS "Riders_Active_2021_2024"
3 FROM riders_raw r
4 JOIN rides_raw rd ON r."Rider_ID" = rd."Rider_ID"
5 WHERE EXTRACT(YEAR FROM r."Signup_Date") = 2021
6       AND EXTRACT(YEAR FROM rd."Pickup_Time") = 2024
7       AND rd."Status" = 'Completed';
8
```

Data Output Messages Notifications

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	Rider_ID integer	Driver_Name character varying (100)	Rider_Name character varying (100)	Pickup_City character varying (100)	Dropoff_City character varying (100)	Distance_KM numeric (10,2)	Payment_Method character varying (50)
1	292	Driver_1774	Rider_292	Calgary	Los Angeles	30.00	Voucher
2	6131	Driver_1886	Rider_6131	Ottawa	Vancouver	30.00	Voucher
3	8405	Driver_1000	Rider_8405	Chicago	San Francisco	29.99	PayPal
4	3294	Driver_1406	Rider_3294	New York	Toronto	29.99	Cash
5	229	Driver_1976	Rider_229	Toronto	Toronto	29.99	Cash
6	7646	Driver_1093	Rider_7646	Calgary	Boston	29.99	Cash
7	2993	Driver_862	Rider_2993	Los Angeles	Toronto	29.99	Cash
8	1599	Driver_1379	Rider_1599	Boston	San Francisco	29.99	Card
9	4175	Driver_1608	Rider_4175	Toronto	Vancouver	29.99	Card
10	9364	Driver_1574	Rider_9364	Boston	Toronto	29.99	PayPal
Total rows: 10		Query complete 00:00:01.774				CRLF Ln 1, Col 80	

Question 3: Quarterly Revenue Comparison and YoY Growth

Objective: Evaluate revenue performance across 2021–2024 and identify quarters with the highest Year-over-Year (YoY) growth.

Findings & Insights:

- Q2 2022 had the highest YoY growth: 200.90%, with \$104,654.66 revenue.
- Reflects strong user acquisition and operational efficiency post-launch.

Query

Query History

1

/* Question 3: Compare quarterly revenue between 2021, 2022, 2023, and 2024.

2

Which quarter had the biggest Year-over-Year (YoY) growth? */

3

SELECT

4

EXTRACT(YEAR FROM r."Pickup_Time") AS "Year",

5

EXTRACT(QUARTER FROM r."Pickup_Time") AS "Quarter",

6

SUM(p."Amount") AS "Total_Revenue"

7

FROM rides_raw r

8

JOIN payment_raw p ON r."Ride_ID" = p."Ride_ID"

9

WHERE p."Amount" > 0

10

AND r."Status" = 'Completed'

11

AND r."Pickup_Time" BETWEEN '2021-06-01' AND '2024-12-31'

12

GROUP BY EXTRACT(YEAR FROM r."Pickup_Time"), EXTRACT(QUARTER FROM r."Pickup_Time")

13

ORDER BY "Year", "Quarter";

14

15

WITH quarterly_yoy AS (

16

SELECT

17

/* ... */

Data Output

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	Rider_ID integer	Driver_Name character varying (100)	Rider_Name character varying (100)	Pickup_City character varying (100)	Dropoff_City character varying (100)	Distance_KM numeric (10,2)	Payment_Method character varying (50)
1	292	Driver_1774	Rider_292	Calgary	Los Angeles	30.00	Voucher
2	6131	Driver_1886	Rider_6131	Ottawa	Vancouver	30.00	Voucher
3	8405	Driver_1000	Rider_8405	Chicago	San Francisco	29.99	PayPal
4	2204	Driver_1406	Rider_2204	New York	Toronto	20.00	Cash

Total rows: 10

Query complete 00:00:01.774

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Question 4: Average Monthly Rides per Driver

Objective: Assess driver consistency and identify top performers by rides per active month.

Findings & Insights:

- Top 5 consistent drivers:

1. Driver_1232 – 1.82 rides/month
2. Driver_138 – 1.70 rides/month
3. Driver_431 – 1.69 rides/month
4. Driver_235 – 1.67 rides/month
5. Driver_537 – 1.67 rides/month

- These drivers maintained steady engagement, showing reliability.

Query

Query History

```
1  /* Question 4: For each driver, calculate their average monthly rides since signup.
2  Who are the top 5 driver with the highest consistency (most rides per active month)? */
3  WITH driver_monthly_rides AS (
4      SELECT
5          r."Driver_ID",
6          TO_CHAR(r."Pickup_Time", 'YYYY-MM') AS "Month_Year",
7          COUNT(r."Ride_ID") AS "Total_Rides"
8      FROM rides_raw r
9      WHERE r."Status" = 'Completed'
10     GROUP BY r."Driver_ID", "Month_Year"
11 )
12
13 SELECT
14     d."Driver_Name",
15     dm."Driver_ID",
16     ROUND(
17         CAST(SUM(dm."Total_Rides") AS NUMERIC) /
```

Data Output

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	Rider_ID integer	Driver_Name character varying (100)	Rider_Name character varying (100)	Pickup_City character varying (100)	Dropoff_City character varying (100)	Distance_KM numeric (10,2)	Payment_Method character varying (50)
1	292	Driver_1774	Rider_292	Calgary	Los Angeles	30.00	Voucher
2	6131	Driver_1886	Rider_6131	Ottawa	Vancouver	30.00	Voucher
3	8405	Driver_1000	Rider_8405	Chicago	San Francisco	29.99	PayPal
4	2204	Driver_1406	Rider_2204	New York	Toronto	20.00	Cash

Total rows: 10

Query complete 00:00:01.774

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Question 5: Cancellation Rate per City

Objective: Detect cities with high cancellation rates to identify operational challenges.

Findings & Insights:

- Highest cancellations: Chicago (19.26%), Toronto (19.08%), San Francisco (18.47%).
- Likely linked to peak-hour demand and driver availability gaps.

[Query](#) [Query History](#)

```
1  /* Qestion 5: Calculate the cancellation rate per city and identify which city had
2  the highest cancellation rate? */
3  SELECT
4      r."Pickup_City" AS "City",
5      ROUND(
6          (SUM(CASE WHEN r."Status" = 'Cancelled' THEN 1 ELSE 0 END)::NUMERIC * 100.0 / COUNT(*)),
7          2
8      ) AS "Cancellation_Rate"
9  FROM rides_raw r
10 GROUP BY r."Pickup_City"
11 ORDER BY "Cancellation_Rate" DESC;
```

[Data Output](#) [Messages](#) [Notifications](#)

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	Rider_ID integer	Driver_Name character varying (100)	Rider_Name character varying (100)	Pickup_City character varying (100)	Dropoff_City character varying (100)	Distance_KM numeric (10,2)	Payment_Method character varying (50)
1	292	Driver_1774	Rider_292	Calgary	Los Angeles	30.00	Voucher
2	6131	Driver_1886	Rider_6131	Ottawa	Vancouver	30.00	Voucher
3	8405	Driver_1000	Rider_8405	Chicago	San Francisco	29.99	PayPal
4	3294	Driver_1406	Rider_3294	New York	Toronto	29.99	Cash
5	229	Driver_1976	Rider_229	Toronto	Toronto	29.99	Cash
6	7646	Driver_1093	Rider_7646	Calgary	Boston	29.99	Cash
7	2993	Driver_862	Rider_2993	Los Angeles	Toronto	29.99	Cash
8	1599	Driver_1379	Rider_1599	Boston	San Francisco	29.99	Card
Total rows: 10		Query complete 00:00:01.774				CRLF Ln 2, Col 34	

Question 6: Riders With More Than 10 Rides But Never Paid Cash

Objective: Understand non-cash payment adoption among active riders.

Findings & Insights:

- 7 riders identified (e.g., Rider_1531, Rider_2672, Rider_3890) with 11–12 rides each, all non-cash.
- Suggests readiness for cashless incentive programs.

Query Query History

```
1 -- Question 6: Identify riders who have taken more than 10 rides but never paid with cash.
2 SELECT
3     r."Rider_ID",
4     rd."Rider_Name" AS "Rider_Name",
5     COUNT(DISTINCT r."Ride_ID") AS "Total_Rides"
6 FROM rides_raw r
7 JOIN payment_raw p
8     ON r."Ride_ID" = p."Ride_ID"
9 JOIN riders_raw rd
10    ON r."Rider_ID" = rd."Rider_ID"
11 WHERE p."Payment_Method" != 'Cash'
12 GROUP BY r."Rider_ID", rd."Rider_Name"
13 HAVING COUNT(DISTINCT r."Ride_ID") > 10;
```

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	Rider_ID integer	Driver_Name character varying (100)	Rider_Name character varying (100)	Pickup_City character varying (100)	Dropoff_City character varying (100)	Distance_KM numeric (10,2)	Payment_Method character varying (50)
1	292	Driver_1774	Rider_292	Calgary	Los Angeles	30.00	Voucher
2	6131	Driver_1886	Rider_6131	Ottawa	Vancouver	30.00	Voucher
3	8405	Driver_1000	Rider_8405	Chicago	San Francisco	29.99	PayPal
4	3294	Driver_1406	Rider_3294	New York	Toronto	29.99	Cash
5	229	Driver_1976	Rider_229	Toronto	Toronto	29.99	Cash
6	7646	Driver_1093	Rider_7646	Calgary	Boston	29.99	Cash

Total rows: 10 Query complete 00:00:01.774 CRLF Ln 1, Col 15

Question 7: Top 3 Drivers by City (Revenue)

Objective: Identify top-earning drivers per city between June 2021 – Dec 2024.

Findings & Insights:

- Example – Boston top drivers:
 1. Driver_1176 – \$474.06
 2. Driver_69 – \$441.31
 3. Driver_815 – \$394.57
- These drivers significantly contributed to city-level revenue.

Query

Query History

```
1 /* Question 7: Find the top 3 drivers in each city by total revenue earned between June 2021 and Dec 2024.
2 If a driver has multiple cities, count revenue where they picked up passengers in the city.*/
3 SELECT
4     r."Pickup_City" AS "City",
5     d."Driver_ID",
6     d."Driver_Name" AS "Driver_Name",
7     SUM(p."Amount") AS "Total_Revenue",
8     RANK() OVER (PARTITION BY r."Pickup_City" ORDER BY SUM(p."Amount") DESC) AS "Driver_Rank"
9 FROM rides_raw r
10 JOIN drivers_raw d ON r."Driver_ID" = d."Driver_ID"
11 JOIN payment_raw p ON r."Ride_ID" = p."Ride_ID"
12 WHERE p."Amount" > 0
13     AND r."Pickup_Time" BETWEEN '2021-06-01' AND '2024-12-31'
14 GROUP BY r."Pickup_City", d."Driver_ID", d."Driver_Name"
15 HAVING SUM(p."Amount") > 0
16 ORDER BY r."Pickup_City", "Driver_Rank"
17 LIMIT 3;
```

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	Rider_ID integer	Driver_Name character varying (100)	Rider_Name character varying (100)	Pickup_City character varying (100)	Dropoff_City character varying (100)	Distance_KM numeric (10,2)	Payment_Method character varying (50)
1	292	Driver_1774	Rider_292	Calgary	Los Angeles	30.00	Voucher
2	6131	Driver_1886	Rider_6131	Ottawa	Vancouver	30.00	Voucher
3	8405	Driver_1000	Rider_8405	Chicago	San Francisco	29.99	PayPal

Total rows: 10 Query complete 00:00:01.774 CRLF Ln 17, Col 9

Question 8: Top 10 Drivers Eligible for Bonuses

Objective: Determine drivers meeting bonus eligibility: ≥ 30 rides, average rating ≥ 4.5 , cancellation $< 5\%$.

Findings & Insights:

- Only 2 drivers met criteria:

1. Driver_1005 – 34 rides, 4.80 rating, 0% cancellations, \$1,462.54 revenue

2. Driver_1181 – 32 rides, 4.60 rating, 3.13% cancellations, \$1,381.92 revenue

- Stringent standards limit qualification, highlighting potential review of bonus thresholds.

Query

Query History

```

1  /* Question 8: Management wants to know the top 10 drivers that are qualified to receive bonuses
2  using the criteria below;•at least 30 rides completed,
3  an average rating ≥ 4.5, and
4  a cancellation rate under 5%.*/
5  SELECT
6      d."Driver_ID",
7      d."Driver_Name" AS "Driver_Name",
8      COUNT(r."Ride_ID") AS "Total_Rides",
9      ROUND(AVG(d."Rating"), 2) AS "AVG_Rating",
10     ROUND(
11         (SUM(CASE WHEN r."Status" = 'Cancelled' THEN 1 ELSE 0 END)::NUMERIC / COUNT(r."Ride_ID")) * 100,
12         2
13     ) AS "Cancellation_Rate",
14     SUM(R."Fare") AS "Total_Revenue"
15 FROM rides_raw r
16 JOIN drivers_raw d ON R."Driver_ID" = d."Driver_ID"
17 WHERE r."Pickup_Time" BETWEEN '2021-01-01' AND '2024-12-31'

```

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	Driver_ID [PK] integer	Driver_Name character varying (100)	Total_Rides bigint	AVG_Rating numeric	Cancellation_Rate numeric	Total_Revenue numeric
1	1005	Driver_1005	34	4.80	0.00	1462.54
2	1181	Driver_1181	32	4.60	3.13	1381.92

Total rows: 2

Query complete 00:00:03.007

CRLF

Ln 5, Col 1

4. Recommendations

- Investigate high cancellations in Chicago and Toronto.
- Consider cashless incentives for riders with consistent non-cash usage.
- Reassess bonus criteria to encourage more drivers while maintaining quality.
- Monitor top-performing drivers to retain talent and reward consistency.

5. Conclusion

- Platform shows steady operational maturity post-2021.
- Revenue and rider retention indicate long-term loyalty and sustainable growth.
- High digital payment adoption presents opportunities for innovation and cashless incentives.
- Driver and city-level insights can guide targeted operational improvements.